12 POINTERS FOR INTERNET2 REAL-TIME ELECTROACOUSTIC COMPOSITION/PERFORMANCE

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ARTIST'S TALK

Using internationally linked high-speed broadband research networks (both IPv4 and IPv6) allows multiple bi-directional audio, video, OSC data, and graphics channels for real-time electroacoustic music composition/performance synchronously. Based on extensive work over the last three years from a research, performance and composition perspective (see Fields & Whalley 2012, for example), the presentation looks at twelve useful pointers others might consider when venturing into this area (Whalley 2012), as electroacoustic music studios become increasing connected real-time globally. This covers:

1. SW/HW - How to get connected.

The use of JackAudio/JackTrip to route audio, OSC Groups for data, video conferencing units for visuals. High-speed research networks such as KAREN and AANET, static IP addresses, firewalls and IPv4/IPv6, and the need for routing and internal testing.

2. Production Space.

Creative Goals (Weinberg 2005) influence the systems used and built for a work – extremes being musical structure/composer controlled, and a processes-based approach. A structure-centred approach allows participants (agents and/or humans) to fill prescribed musical or performance outcomes. A process-centred approach explores possibilities or fulfills goals through collaboration or competition, and experiences may differ with each session.

3. Network Arrangement.

For the exchange of real-time audio two useful models are: a Peer to Peer Approach that has no central sound server - each peer streams to close peers making a cluster network, and each peer transmits many and receives many streams with no central mixing. A Star Approach provides a central server that collects individual outputs and sends out individual streams. Peers then transmit one stream and receive many, and they can alter individual streams

4. Latency and distances.

Latency cannot be avoided, and the choices are cope accept it, ignore it, or play with it (Carôt, Rebelo, Renaud, 2007). Synchronously, the two choices available they put forward are a Realistic Jam Approach (RJA) and a Latency Accepting Approach (LAA). LAA is more suited to electroacoustic music, where one learns to compose/improvise in ways that timing is not critical, and/or use strategies to compensate for it in aesthetic terms. Latency/delay is also a creative opportunity, such as using network delays as multiple musical delays.

5. Production Integration

Follmer (2005) sets out a three dimensional matrix to assess approaches to the NMP idiom that can include Internet2 practice. The matrix maps: Interplay with Network Characteristics – how the structural characteristics of the network shape the work. Interactivity and Openness or the extent that users/audiences can join in with and influence outcomes. Complexity and Flexibility – or how musically/sonically flexible the outcome can be. Follmer suggests that the production approaches best suited to the medium are hypermusic (new hybrid instruments); real-virtual/space sound installations; and algorithmic/generative installations.

6. Linear/non-liner shift and communication.

In general, linear/edited electroacoustic music has often sacrificed spontaneity and interactivity for the sake of deterministic control of increasing Parameter Space, where time is secondary to innovation and reflection toward perfect product. Interactive and generative systems often sacrifice aspects of real-time Parameter Space in favour of spontaneous outcomes that can be multiple and varied.

7. Relationships.

Roles and Motivation**.** Considering motivates people to want to invest in and take part in telemusic events in a cost/benefit sense is vital, given that in contrast to on-site music making methods it is often complicated, expensive to assemble, and more time consuming. Access to Internet2 is also limited globally to a few, and infrastructure still favours partners in closest physical proximity to each other because of time delays.

8. Motivation, net access, payoff.

Why do it? It allows the forging of new relationships that can be artistically rewarding beyond what could be produced by other means, but that these relationships take time to build through rehearsal processes. In tandem, the sense of playing simultaneously in both real and virtual environments can be rewarding in the sense of extending musical proximity through co-located action.

9. Dramaturgy models in Internet2. Relationships models (Rebelo 2009) based in traditional performance practice to get things done through Internet2: Projected Dramaturgy - one node to act as the author and other nodes to contribute or project to it; Directed Dramaturgy - authorship remains with a group or individual who takes on a director's role. Distributed Dramaturgy - where authorship is retained by each node, while contributing expertise and content to shared production.

10. Relationship: machine role considerations

Given non-trivial machines and OSC messages can be exchange between individuals/machines and between groups through Internet2, the role of intelligent machines implemented through multi-agent technology needs considering (Whalley 2009). Practitioners have two choices at or between nodes include: a multi-agent system that is self-contained (generative model), or one that can include external input (machine or human) to make a generative improvisational model using intelligent machines

11. Instrument choices.

Beyond just adding effects to acoustic instruments as new hybrids, telemusic has its own aesthetics, and needs to draw on instrument building traditions suited to this environment, and optimize performance for it. This extends the LAN centric laptop orchestra models exploring Internet2.

12. Changing roles: makers/machines/audiences.

The new space anticipates extending current interactive music paradigms, changing the traditional roles of composer, musician and producer. It also anticipates the creative role programmers are likely to play in Internet2 works, the further deployment of intelligent machines, and more participatory roles for audiences.

Extending current work using Video Conferencing Units (VCU) as one part of linked performances, our current work with international partners explores a means of distributing graphics scores that can be altered in real-time across multiple countries over a fourth network channel in addition of VCU, OSC Groups, and JackTrip Audio. This investigates what is useful from both a pragmatic performance perspective, and what is useful as a means to enhance the visual aspects of live performance layers in electroacoustic music. This is because all collated performance outcomes can also be streamed to the public Internet in real-time, extending the worldwide audience for outcomes in this new medium.

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